

CS-Mamba: A Novel Mamba-Based Framework for Landslide Detection With Attention Guided Cross-Scale Context Semantic Integration

Yanrong Mao , Haiming Zhang , Rongfan Dai , Tingjian Yang, and Guanting Liu

Abstract—Landslides are a frequent natural hazard, and the precise detection of affected areas using high-resolution optical imagery is of great significance for geological disaster emergency response and risk assessment. In optical images, landslides often exhibit substantial scale variations, complex and diverse morphologies, and blurred boundaries, and are susceptible to environmental influences. Existing deep learning approaches have not fully exploited cross-scale dependencies among pixels or ensured the consistency between feature representations and semantic embeddings, leading to insufficient model robustness. To address these limitations, we propose a novel Mamba-based framework for landslides detection with attention-guided cross-scale context semantic integration (CS-Mamba). In the encoder, parallel multiscale convolutional branches are introduced to enhance the network's representation capability. At the core, a multiscale-aware state-space module in the neck adopts bidirectional information flows and a gated fusion mechanism to integrate global and local features, thereby improving semantic consistency in boundary segmentation. In the decoder, an attention-guided strategy is employed to fuse multilevel features, achieving precise preservation of boundary details and strong robustness to cross-domain data. The proposed method is evaluated on multiple representative landslide datasets, including the large-scale multisource high-resolution landslide dataset, Bijie, and GLVM datasets. Experimental results demonstrate that CS-Mamba achieves superior performance on nearly all metrics and exhibits strong generalization ability in domain generalization analysis.

Index Terms—Deep learning, high-resolution remote sensing imagery, Landslide detection.

I. INTRODUCTION

LANDSLIDES are a common and highly destructive natural hazard that poses serious threats to human life, property, and infrastructure. Influenced by the intensification of global climate change and the acceleration of urbanization, the frequency and severity of landslide events have been continuously increasing [1]. According to statistics released by the National Bureau of Statistics of China, a total of 2335 landslide events occurred in 2024, resulting in 129 casualties, including 8 fatalities, and causing direct economic losses of up to CNY 3.20 billion [2]. Consequently, the timely and accurate detection of affected areas after a landslide has become a pressing challenge in response to emergency geological disasters and risk assessment.

Traditional landslide detection methods are largely based on manual visual interpretation, typically involving field surveys and expert experience for comprehensive analysis and judgment. However, such approaches are highly dependent on expert knowledge, have low detection efficiency, and are difficult to implement on a large scale. In addition, manual interpretation is prone to omissions and false detections, making it difficult to ensure the accuracy and reliability of the results. With the rapid development of high-resolution remote sensing technology, landslide detection methods based on high-resolution optical imagery acquired from satellite and airborne platforms have increasingly become a research focus [3], [4], [5].

In recent years, deep learning has shown strong advantages in landslide detection due to its end-to-end modeling, feature learning ability, and high level of automation [6]. Its performance, particularly in handling large-scale, high-resolution remote sensing imagery, has significantly surpassed that of traditional machine learning methods [7]. Many high-quality landslide datasets have been built from satellite, airborne and UAV imagery at different scales [3], [8], [9], [10], [11]. For example, the global high-resolution landslide dataset constructed by Liu et al. [11] integrates diverse remote sensing data sources covering various terrains, triggering factors, and imaging platforms, greatly advancing research on landslide detection methods. Despite the growing availability of data and continuous improvement in model performance, landslide detection still faces multiple challenges, especially in multisource data fusion and cross-domain generalization. As shown in Fig. 1, landslide detection is challenged by the scale difference among landslides, the imaging condition discrepancies across remote sensing platforms, and the

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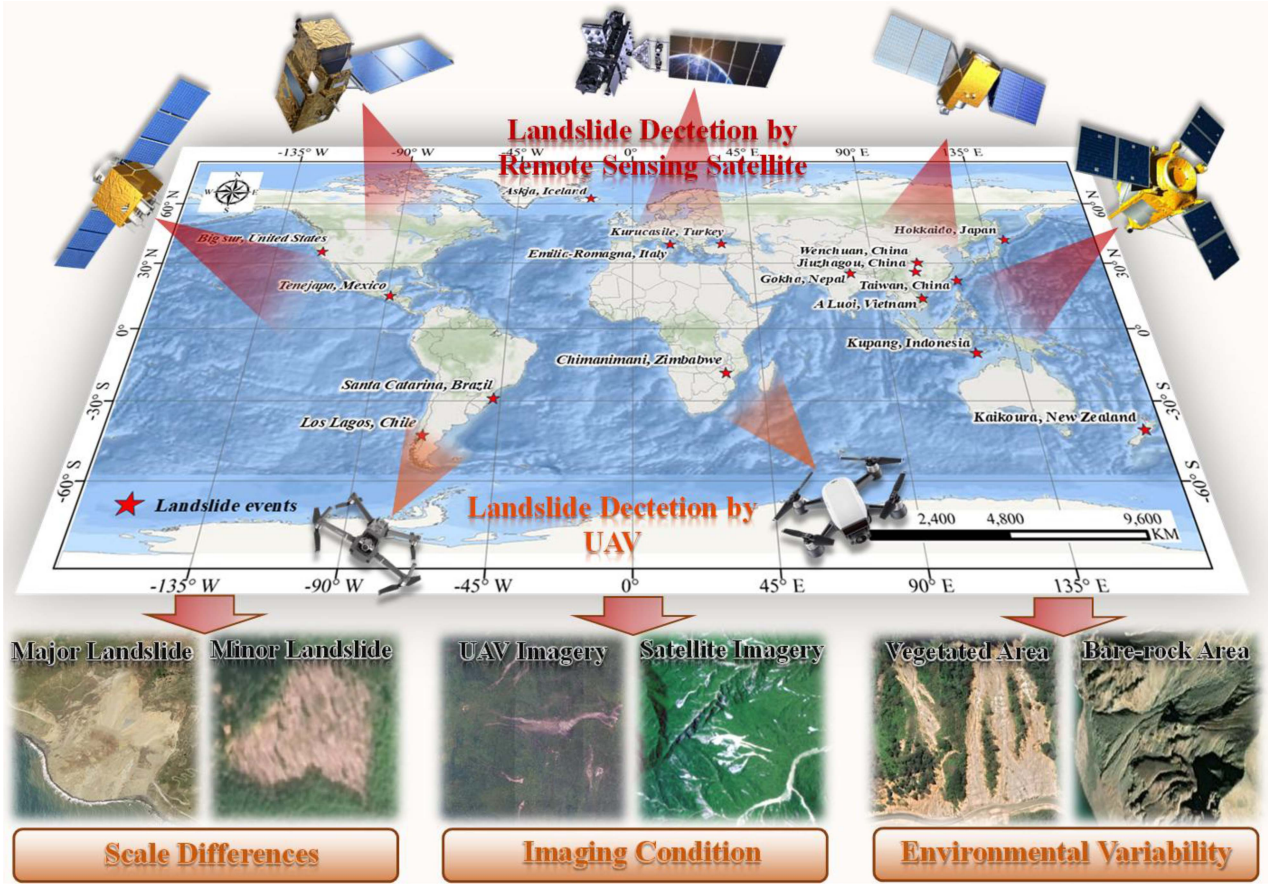


Fig. 1. Global distribution of landslide and major challenges in landslide detection. Sixteen representative landslide sites worldwide are marked based on publicly available datasets.

environmental variability of occurrence regions. These factors cause landslides to exhibit large differences in size, texture, color, and boundary features across images, resulting in substantial accuracy degradation when existing models are applied to new regions or data sources. Therefore, developing a landslide detection model with strong adaptability and generalization capability across scales, imaging conditions, and environments has become a key research issue [12].

The convolutional neural networks (CNNs) have been widely applied to landslide detection, such as Mask R-CNN [13], ResU-Net [14], YOLO [15], SCGC-Net [16], but their inherent local receptive fields limit their ability to model long-range dependencies. Furthermore, repeated downsampling inevitably degrade fine-grained boundary information, while conventional upsampling and skip connections often prove insufficient for restoring precise edges [17]. Additionally, the lack of explicit cross-scale semantic coordination leads to inconsistencies between high-level semantic predictions and low-level spatial details, particularly for landslides with complex geometries or small spatial extents [18]. The self-attention mechanism introduced by Transformer architectures offers strong global modeling capability, yet suffers from high computational costs, sensitivity to dataset size, and a tendency to overlook fine-grained boundary information [19], [20], [21], the global attention mechanism tends to prioritize regional-level semantic consistency, which may

smooth out high-frequency boundary cues and consequently exacerbate boundary ambiguity in landslide segmentation tasks. Recently, the Mamba state space model (SSM) has emerged as a promising alternative. By leveraging state space modeling, Mamba can efficiently capture long-range dependencies while maintaining linear time complexity [22], and has demonstrated strong performance in vision tasks including image classification, object detection, and segmentation [23], [24], [25], [26]. However, existing Mamba-based vision models primarily focus on efficient long-range modeling and often lack specialized mechanisms for multiscale semantic alignment and boundary-aware feature fusion. This limitation restricts their effectiveness in complex landslide scenarios, and their application in landslide detection remains in its nascent stage.

To address the aforementioned issues, this article proposes a novel landslide detection framework (CS-Mamba) designed to achieve comprehensive multiscale contextual integration, precise preservation of boundary details, and strong robustness to cross-domain data. The CS-Mamba architecture comprises three core modules.

- 1) Cross embedding module (CEM), which fuses features at different scales to enhance the representation of landslide morphology.
- 2) Multiscale aware visual state space (MAVSS) block, which combines the strengths of Mamba and CNNs by

introducing bidirectional information flow across different feature levels and employing a gated fusion mechanism to integrate global and local features, thereby improving segmentation accuracy and semantic consistency along boundaries.

- 3) Semantic fusion module (SFM), which applies an attention-guided strategy to integrate multilevel features, bridging the gap between global context and local details, and dynamically adjusting features across layers to improve segmentation performance for blurred or fragmented landslide regions.

CS-Mamba was systematically evaluated on three landslide datasets. The results show that it outperforms existing mainstream methods on multiple metrics such as F1-score and mIoU, achieving the highest mIoU of 86.18% on the Bijie dataset [8], 90.90% on the global very-high-resolution landslide mapping (GVLM) dataset [10], and 87.11% on the large-scale multisource high-resolution landslide dataset (LMHLD) [11]. It also demonstrates excellent robustness in cross-domain testing, requiring only a small number of samples to adapt the model to previously unseen datasets. In summary, the proposed CS-Mamba provides a high-performance, efficient, and highly generalizable solution for high-resolution remote sensing-based landslide detection, offering significant theoretical value and practical potential for the development of intelligent geological hazard monitoring systems.

II. RELATED WORK

A. Landslide Detection in Deep Learning

Early landslide detection models often employed CNN, which stack convolution and pooling layers followed by fully connected layers [7], [27], [28]. Subsequently, Ji et al. [8] introduced a novel attention module to develop a more complex and specialized CNN architecture and constructed an optical landslide dataset, filling a gap in the field of landslide detection. Yi and Zhang [29] further investigated the application of CNN to optical imagery-based landslide detection and proposed LandsNet, a cascaded end-to-end CNN. To enhance multiscale modeling capabilities, Liu et al. [30] replaced the original bilinear interpolation in the decoder with channel attention mechanisms and atrous spatial pyramid pooling. Considering the instability of existing landslide detection models in complex scenarios and their limited boundary delineation accuracy, Li et al. [31] designed a contrastive learning-guided landslide detection model with high robustness. By incorporating residual blocks and channel attention mechanisms into the U-Net+++ architecture, they achieved significant improvements in boundary segmentation accuracy. Wang et al. [32] innovatively designed a gated dual-stream CNN that largely mitigated information loss during deep downsampling; however, the intrinsic local receptive field of CNN still restricts their ability to model long-range dependencies, limiting attention to global contextual relationships.

The emergence of the Transformer architecture has provided an alternative solution. To address the wide-ranging terrain variations and morphological diversity of landslides in remote sensing imagery, Lv et al. [20] designed the Shape-Enhanced

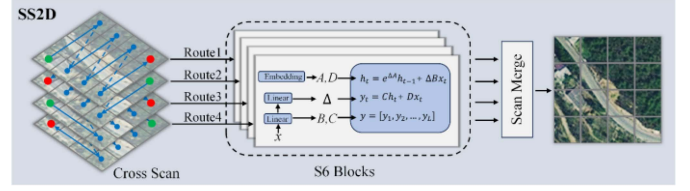


Fig. 2. 2D-selective-scan (SS2D).

Vision Transformer (ShapeFormer) to more accurately capture and preserve multiscale shape features of landslides. Similarly, Huang et al. [33] combined morphological edge detection with Swin-Transformer to further improve the accuracy of landslide shape and boundary extraction. To leverage the efficiency of CNN in local feature extraction and the strength of transformers in global dependency modeling, researchers have explored hybrid CNN–Transformer architectures to achieve complementary benefits. Li et al. [34] coupled CNN and Transformer in a multiscale feature fusion scene parsing framework to link the rich features of landslides, enhancing model expressiveness. Wu et al. [35] integrated shallow local features from CNN with deep global features from Transformers through multistage fusion, effectively improving cross-scale feature representation and boundary delineation accuracy in landslide detection.

Despite their superior performance compared with single-architecture models, CNN–Transformer hybrids still have limitations: the Transformer module entails high computational and memory costs, which can create resource bottlenecks when processing high-resolution remote sensing imagery; furthermore, the feature fusion between CNN and Transformers often relies on manually designed mechanisms and lacks efficient modeling of cross-scale sequential dependencies. The recently proposed Mamba architecture [22] offers a new research direction for remote sensing applications such as landslide detection. Based on the SSM, Mamba achieves long-sequence modeling with linear computational complexity, overcoming the quadratic complexity constraint of global attention in Transformers.

B. Mamba Architecture

The Mamba architecture, proposed by Gu and Dao [22], is built upon the SSM and aims to achieve global modeling of long sequences with linear time complexity. The fundamental concept of SSM is to represent the dynamic relationship between the input sequence and the hidden states using continuous-time state equations, which can be expressed as follows:

$$h'(t) = Ah(t) + Bx(t), y(t) = Ch(t) + Dx(t) \quad (1)$$

where $h(t)$ denotes the hidden state, while $x(t)$ and $y(t)$ represent the input and output, respectively. The matrices A , B , C , and D are learnable parameters. Building upon the conventional SSM, Mamba incorporates a selective mechanism, in which the state-transition parameters are dynamically generated during computation. This design enables the model to adaptively regulate the proportion of information retained and forgotten according to the input content. Such a mechanism substantially

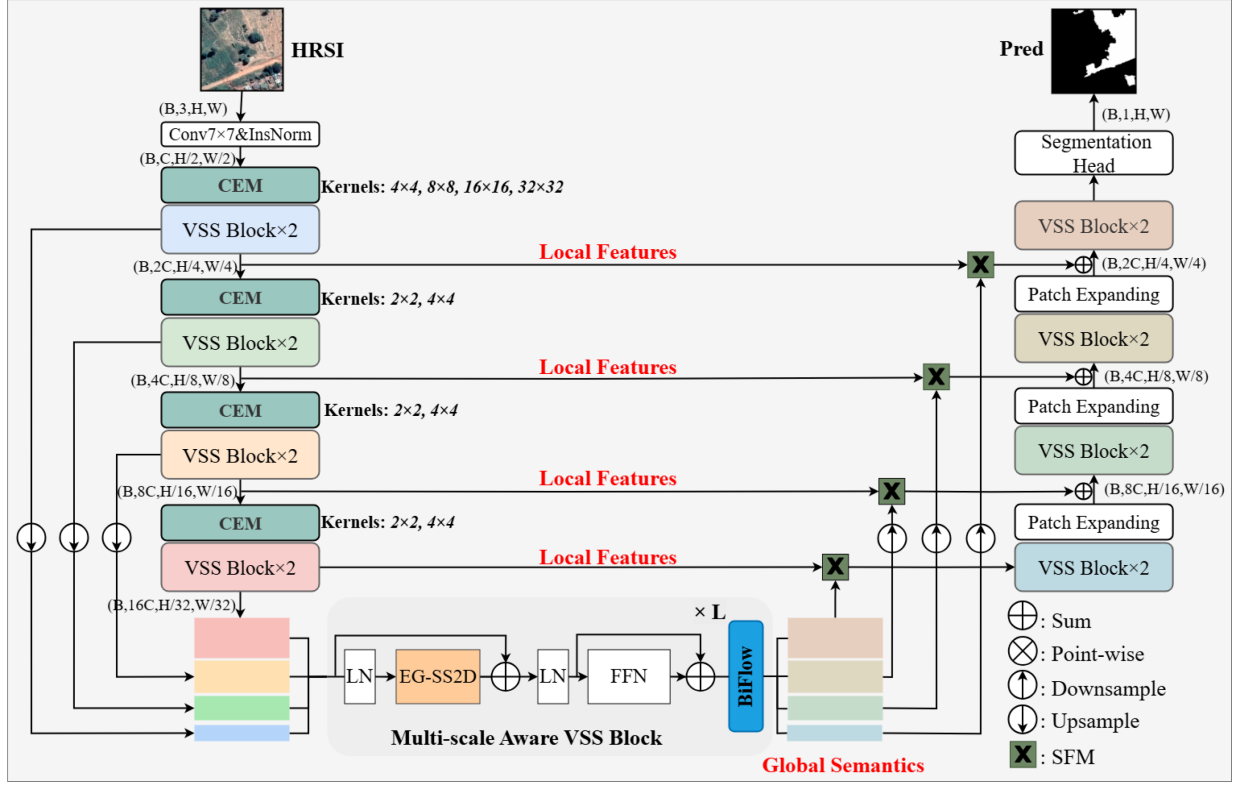


Fig. 3. Architecture of the proposed CS-Mamba.

enhances the granularity of information selection and the capability for modeling long-range dependencies, while maintaining linear growth in both computational cost and memory usage.

The original Mamba model was designed for one-dimensional sequence tasks, whereas VMamba [24] extends it to the computer vision domain. At its core lies the visual SSM, which incorporates the 2D selective scan (SS2D) to enable global dependency modeling of two-dimensional images while retaining linear computational complexity. Since image features in the 2D spatial domain lack a strict sequential order, but require capturing contextual information along both horizontal and vertical directions, VMamba introduces a four-directional scanning strategy within SS2D, as illustrated in Fig. 2.

III. PROPOSED METHOD

This section primarily presents the overall architecture of CS-Mamba. As shown in Fig. 3, the model is inspired by the U-Net architecture and consists of three main modules: CEM, it explicitly align semantic representations across different spatial scales, mitigating semantic inconsistency caused by large-scale variations of landslides. MAVSS block, it serves as the core component of the network, jointly modeling long-range global dependencies and fine-grained local details to improve semantic consistency and boundary delineation. SFM, it is applied in the decoding stage to adaptively integrate global semantics into multiscale feature maps, further refining ambiguous boundaries and fragmented regions. Finally, a lightweight segmentation head predicts the landslide mask at the original resolution.

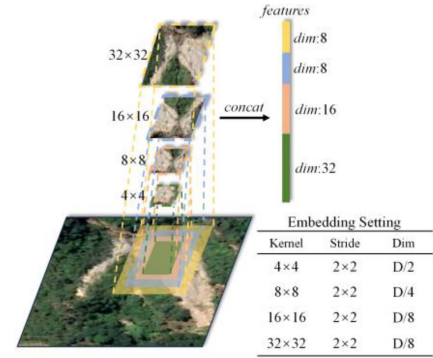


Fig. 4. Cross embedding module.

The subsequent sections provide a detailed description of each module and the corresponding data flow mechanisms.

A. Cross Embedding Module

To address the challenge of cross-scale semantic inconsistency caused by large variations in landslide size, the encoding stage of CS-Mamba employs a CEM as the initial feature extraction unit. The core function of CEM is to encode landslide features at multiple spatial scales in a coordinated manner, enabling fine-grained local textures and broad contextual structures to be represented within a unified semantic space. As illustrated in Fig. 4, taking the first encoding stage as an example, the CEM constructs four parallel convolutional branches, each using a distinct kernel size $K \in \{4, 8, 16, 32\}$ and a shared stride setting

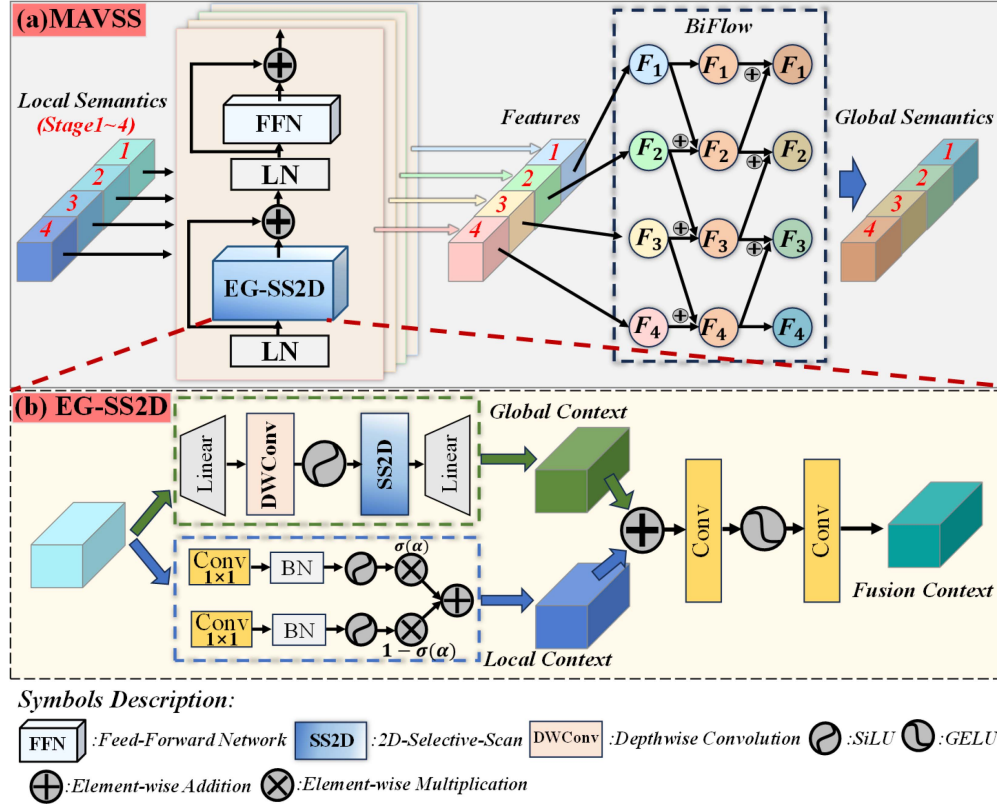


Fig. 5. Multiscale aware visual state space block.

to ensure that the output tensors maintain consistent spatial resolution. Each branch focuses on contextual information at a specific scale, where the sampling regions share the same spatial center, but vary in coverage extent.

To balance computational efficiency and representation capacity, CEM adopts a nonuniform channel allocation strategy, where branches with larger receptive fields are assigned fewer channels. This design effectively controls computational cost while preserving the ability to model multiscale landslide characteristics. Similar CEM structures are applied at different encoding stages, with kernel sizes and stride settings adapted to the network's downsampling process to form a pyramid-shaped semantic hierarchy.

B. Multiscale Aware Visual State Space Block

The MAVSS block is the key module of CS-Mamba, designed to address cross-scale semantic inconsistency and boundary ambiguity in landslide detection. The neck of the model consists of L stacked MAVSS blocks. Each MAVSS block contains an Efficient Global SS2D, a feed-forward neural network, and residual connection.

As illustrated in Fig. 5(b), the efficient global SS2D incorporates both a global branch and a local branch. The global branch uses the modeling mechanism of the vision SSM to capture long-range spatial dependencies and geomorphological semantic features across the image. The local branch adopts depthwise

separable convolutions to enhance the characterization of fine-grained local features such as landslide boundaries. The outputs of the two branches are integrated through a gated weighting mechanism, which further improves the consistency and discriminative capacity of multiscale features. The specific processing of the efficient global SS2D can be expressed as follows:

$$\begin{aligned}
 F_{\text{global}} &= W_2 \cdot \text{LN}(\text{Scan}(\text{DWConv}(W_1 \cdot x))) \\
 F_{\text{local}} &= [1 - \sigma(\alpha)]\text{BN}(\text{Conv1}(W \cdot x)) \\
 &\quad + \sigma(\alpha)\text{BN}(\text{Conv3}(W \cdot x)) \\
 F_{\text{fuse}} &= \text{Conv1}(\text{GELU}(\text{Conv1}(\text{LN}(F_{\text{global}} + F_{\text{local}})))) \quad (2)
 \end{aligned}$$

where W denotes the linear layer weights, LN and BN refer to layer normalization and batch normalization, respectively, Scan represents the SS2D operation, DWConv denotes the depthwise separable convolution, GELU is the activation function, and $\sigma(\alpha)$ is a learnable gating parameter activated by the Sigmoid function.

To further enhance cross-scale feature consistency, MAVSS incorporates a bidirectional cross-scale feature flow mechanism (BiFlow), as shown in Fig. 5(a). In the bottom-up path, high-level features are upsampled and fused with low-level features to supplement global semantic information. In the top-down path, low-level features are downsampled and fused with high-level features to provide fine-grained detail feedback.

The MAVSS module first processes the input multiscale features in parallel, where the EG-SS2D extracts both global and

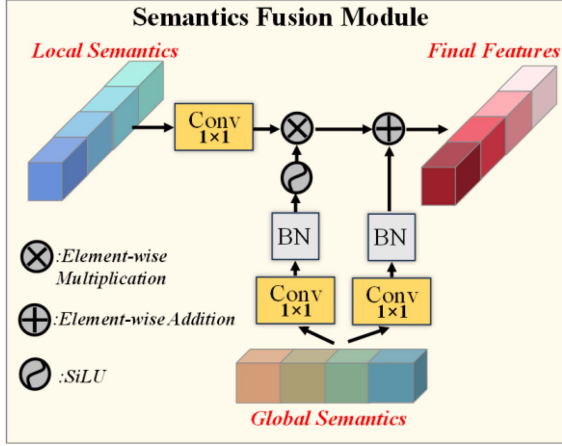


Fig. 6. Semantics fusion module.

local feature representations of the current layer. Subsequently, cross-layer features are dynamically weighted and fused with the current layer features through the bidirectional flow mechanism (BiFlow), which effectively promotes feature complementarity and consistency enhancement across different levels and receptive fields. Through the combination of global-local feature modeling and bidirectional cross-scale interaction, MAVSS produces multilevel representations that preserve fine structural details while maintaining semantic coherence across scales, which is crucial for accurate landslide segmentation under complex geomorphological conditions.

C. Semantic Fusion Module

After obtaining multiscale semantic features, an SFM is introduced in the decoding stage to alleviate boundary ambiguity and bridge the representational gap between local spatial details and global semantics. The SFM takes local features from different encoder levels and global semantic features as inputs, as illustrated in Fig. 6. Local features are first processed by a 1×1 convolution followed by batch normalization to preserve spatial structures and boundary information. In parallel, global features are transformed through two lightweight branches: one branch generates spatial semantic weights via a 1×1 convolution; batch normalization; and Sigmoid activation to adaptively modulate local features, while the other branch produces residual global context to supplement semantic information. The semantic weights are applied to the local features through element-wise multiplication, followed by residual fusion with the global contextual features. This adaptive fusion strategy enables spatially selective enhancement of ambiguous regions while maintaining fine-grained boundary details. After fusion, the enhanced multiscale features are passed to the decoding stage, where they are upsampled, aligned, and integrated to produce accurate landslide segmentation results.

C. Loss Function

To further improve the accuracy of the model in landslide extraction tasks, especially when facing challenges such as large scale variance of landslide samples, severe class imbalance,

and blurred landslide boundaries in the dataset, we design a structure-aware composite loss function. This function dynamically combines the adaptive binary cross-entropy (adaptive BCE) loss and the structure-aware dice loss, enabling simultaneous enhancement of pixel-level classification accuracy and fine-structure segmentation capability. The formulation of the loss function is given as follows:

$$L = \lambda_1 L_{ABCE}(y, \hat{y}) + \lambda_2 L_{SDice}(y, \hat{y}) \quad (3)$$

where y and $\hat{y}$ denote the ground truth and the prediction of the model, respectively, while λ_1 and λ_2 are the weighting coefficients of the two loss functions, which are set to 1 and 0.5 by default.

The L_{ABCE} employs a positive hyperparameter γ to control the difficulty weighting, which adaptively assigns higher weights to hard-to-classify pixels. This design effectively alleviates the issues of class imbalance and hard examples. Its computation is defined as follows:

$$L_{ABCE} = -\frac{1}{N} \sum_{i=1}^N (1 + |y_i - \hat{y}_i|)^\gamma [y_i \log \hat{y}_i + (1 - y_i) \log (1 - \hat{y}_i)] \quad (4)$$

The L_{SDice} applies a gradient operator [36] to process the ground truth y to compute its boundary response g_i , which is used for pixel weighting to strengthen the model's attention to boundary regions. The computation is given as follows:

$$L_{SDice} = 1 - \frac{2 \sum_{i=1}^N (1 + \alpha g_i) y_i \hat{y}_i}{\sum_{i=1}^N y_i + \sum_{i=1}^N \hat{y}_i + \varepsilon} \quad (5)$$

where α denotes a positive hyperparameter, and ε represents a small constant introduced to prevent division by zero.

IV. EXPERIMENT

A. Experimental Configuration

1) *Landslide Datasets*: In this experiment, the proposed model is evaluated on several publicly available landslide datasets, including the Bijie dataset [8], GVLM dataset [10], and LMHLD dataset [11], all of which provide pixel-level annotations for optical high resolution imagery. Fig. 7 presents representative examples from these datasets, highlighting common challenges encountered in optical landslide detection. The red boxes indicate blurred landslide boundaries, the blue boxes represent interference from complex backgrounds, and the yellow boxes denote occlusions caused by shadows. These datasets exhibit substantial diversity in terms of geographic environments, landslide types, and imaging conditions, thereby providing reliable data support for a comprehensive evaluation of the proposed method's performance. The detailed characteristics of these datasets are given in Table I.

The Bijie dataset focuses on Bijie City in the northwest of Guizhou Province, China. It contains 770 landslide samples and 2003 nonlandslide samples collected between May and August 2018. The landslide samples include a total of 7.23×10^6 pixels and comprise rockfalls, rockslides, and several debris slides. The image resolutions range from 61×61 pixels to 1239×1197 pixels.

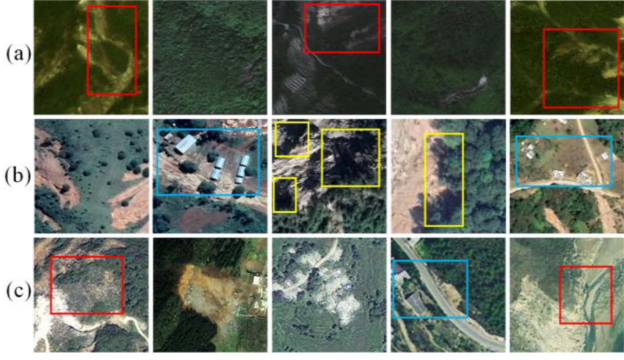


Fig. 7. Landslide datasets and examples of challenges in landslide detection. (a) Examples of LMHLD dataset. (b) Examples of GVLM dataset. (c) Examples of Bijie dataset. The red boxes indicate blurred landslide boundaries, the blue boxes represent interference caused by complex backgrounds, and the yellow boxes denote occlusions resulting from shadows in the imagery.

TABLE I
DIFFERENT LANDSLIDE DATASET STATISTICS

Dataset	Bands	Events	Tiles	Labeling pixels	Patch size
LMHLD	4	7	7	7.23×10^6	128
GVLM	3	17	17	3.24×10^7	512
Bijie Landslides	3	1	2773	5.74×10^7	256

The GLVM dataset consists of 17 distinct landslide sites. It contains bi-temporal images of these sites acquired from Google Earth, with a spatial resolution of up to 0.59 m. The dataset also includes a variety of negative samples, such as nonlandslide terrain, buildings, and transportation infrastructure. The image sizes range from 186×1749 pixels to 10828×7424 pixels. The landslide samples contain a total of 3.24×10^7 pixels.

The LMHLD dataset includes high-resolution landslide imagery from seven regions worldwide, collected using five different satellite sensors. It contains 28185 image patches, with the landslide samples accounting for 5.74×10^7 pixels in total. The landslide pixel counts in individual patches range from as few as one pixel to as many as 325589 pixels, covering a wide range of landslide scales and types.

2) *Evaluation Metrics*: This article employs five commonly used quantitative evaluation metrics to comprehensively assess the performance of the proposed model: precision; recall; mean intersection over union (mIoU); pixel accuracy (PA); and the F1 score. These metrics are also widely adopted in other landslide detection studies [7], [8], [9], [37]. The definitions of these metrics are as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (6)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (7)$$

$$\text{mIoU} = \frac{TP}{TP + FN + FP} \quad (8)$$

$$\text{OA} = \frac{TP + TN}{TP + FP + TN + FN} \quad (9)$$

$$\text{F1} = 2 \times \frac{(\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}. \quad (10)$$

True positive (TP) refers to pixels that are actual landslide pixels and are correctly predicted as landslide pixels by the model. False positive (FP) denotes nonlandslide pixels that are incorrectly classified as landslide pixels. True negative (TN) represents pixels that are correctly identified as nonlandslide pixels, whereas false negative (FN) refers to landslide pixels that are incorrectly classified as nonlandslide pixels, corresponding to missed detections. Precision measures the proportion of correctly predicted landslide pixels among all pixels predicted as landslides, where a higher value indicates better performance in reducing false alarms. Recall evaluates the model's ability to detect actual landslide pixels, with a higher value indicating a lower missed detection rate. Overall accuracy (OA) measures the proportion of correctly classified pixels over the entire image or dataset and serves as a fundamental indicator of overall classification performance.

mIoU quantifies the degree of overlap between the predicted landslide region and the ground truth landslide region, representing an important measure of spatial consistency. The F1 score is the harmonic mean of Precision and Recall and is particularly suitable for comprehensive performance evaluation in scenarios with severe class imbalance, such as when the proportion of landslide pixels is relatively small.

3) *Implementation Details*: All experiments in this article were conducted on an Ubuntu 20.04.5 LTS system, using an NVIDIA Tesla T4 GPU with 16 GB of memory for model training and evaluation. The experiments were implemented with the PyTorch deep learning framework[38]. Different patch sizes were adopted for different datasets to comprehensively assess the flexibility of the model: 128×128 for the LMHLD dataset, 256×256 for the Bijie dataset, and 512×512 for the GVLM dataset.

Each dataset was randomly divided into training, validation, and testing sets following a consistent ratio (8:1:1), ensuring that samples from the same original image were assigned to the same split to avoid spatial overlap between sets. During training, standard data augmentation techniques, including random rotation and scaling, were applied to improve model generalization and robustness to variations in landslide appearance.

In the backward stage, stochastic gradient descent was used as the optimizer with a momentum coefficient of 0.9, an initial learning rate of 0.01, and a weight decay of 0.0005. The learning rate was scheduled using the MultiStepLR strategy, where a decay factor of $\gamma = 0.1$ was applied at the 25th, 35th, and 45th epochs.

B. Comparison With State-of-the-Art Methods

To benchmark the performance of the proposed method and verify its effectiveness, multiple deep learning models with strong performance in visual tasks were selected for comparison experiments on three landslide datasets. These methods include approaches based on conventional CNN architectures as well as emerging Transformer and Mamba architectures. Specifically, the models are FCN [39], UNet [40], ResUNet [41], SegNet [42], DeepLabV3+ [43], Unetformer [44], TransUNet [45], Segformer [46], Vmamba [24], and RS3Mamba [47]. The experiments qualitatively and quantitatively analyze the performance

TABLE II
QUANTITATIVE PERFORMANCE OF VARIOUS METHODS ON LMHLD, GVLM AND BIJIE DATASETS

Method	LMHLD			GVLM			Bijie		
	OA(%)	mIoU(%)	F1(%)	OA(%)	mIoU(%)	F1(%)	OA(%)	mIoU(%)	F1(%)
FCN	92.84	74.22	83.98	98.92	89.68	94.31	95.93	83.45	90.51
UNet	96.73	71.24	80.58	97.29	87.49	93.05	93.49	73.38	83.16
DeepLabV3+	95.47	82.49	89.89	97.79	89.07	94.01	95.92	84.13	90.97
SegNet	95.51	81.80	89.41	97.80	89.25	94.12	95.35	81.92	89.51
ResUNet	95.89	83.89	90.81	98.83	88.96	93.87	91.67	74.78	84.64
UNetformer	95.41	83.29	90.45	98.18	90.35	94.76	95.56	80.42	88.41
TransUNet	94.63	80.68	88.71	97.38	87.59	93.11	92.91	80.14	88.53
Segformer	94.97	80.79	88.74	98.57	90.18	94.64	93.45	77.91	86.81
VMamba	95.41	84.50	90.45	97.92	88.52	94.31	96.47	83.04	90.17
RS3Mamba	94.71	86.15	92.39	95.29	88.25	93.66	95.11	85.41	91.89
Ours	95.63	87.11	92.93	98.88	90.90	95.18	97.92	86.18	92.15

Red indicates the highest performance, followed by orange and green indicating the second and third highest, respectively.

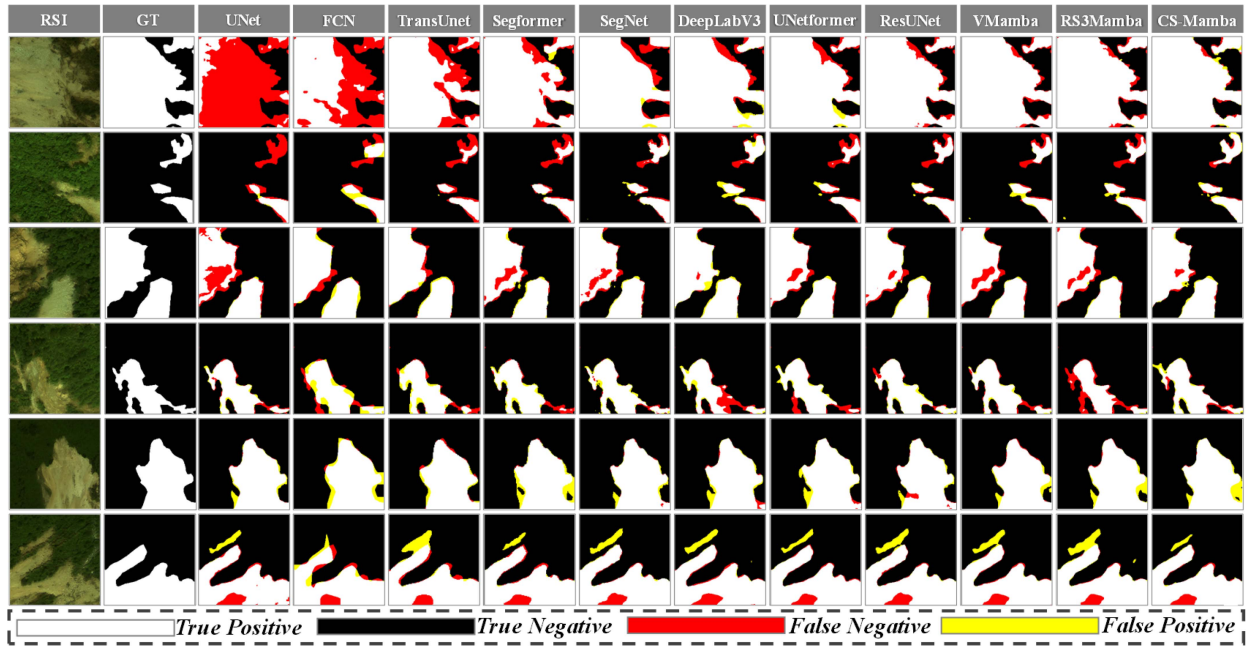


Fig. 8. Visual comparison of landslide detection results produced by different methods on LMHLD dataset.

of CS-Mamba from two perspectives. Table II gives a comprehensive comparison of CS-Mamba and other state-of-the-art methods on the LMHLD, GVLM, and Bijie datasets using three aggregated metrics. The top three results are highlighted in red, orange, and green, respectively, in descending order of performance. To further compare the detection performance of different methods, representative samples from each dataset and their detection results from various models are selected.

On the LMHLD dataset, the proposed CS-Mamba demonstrates a clear advantage in the aggregated evaluation metrics, achieving the best performance with an mIoU of 87.11% and an F1 score of 92.93%, with the mIoU exceeding the third-best result by 2.61%. As shown in Fig. 8, the detection results show higher accuracy in both the boundaries and interior regions of landslides. Compared with other models, CS-Mamba

significantly reduces the red regions representing missed detections, and the segmented patches align closely with the ground truth. In particular, in small landslide patches and areas with complex boundaries, many conventional methods, such as UNet, FCN, and Segformer, tend to miss a substantial number of pixels, producing blurred or even broken boundaries. In contrast, CS-Mamba yields clear boundaries with minimal missed and false detections, indicating that its cross-scale feature flow and semantic fusion mechanisms play a critical role in enhancing boundary restoration and fine-grained recognition.

For the GVLM dataset, CS-Mamba also attains the highest scores in mIoU and F1, reaching 90.90% and 95.18%, respectively. It effectively suppresses false detections in scenes with complex backgrounds and numerous nonlandslide objects, while accurately capturing boundary details. As shown in Fig. 9, the

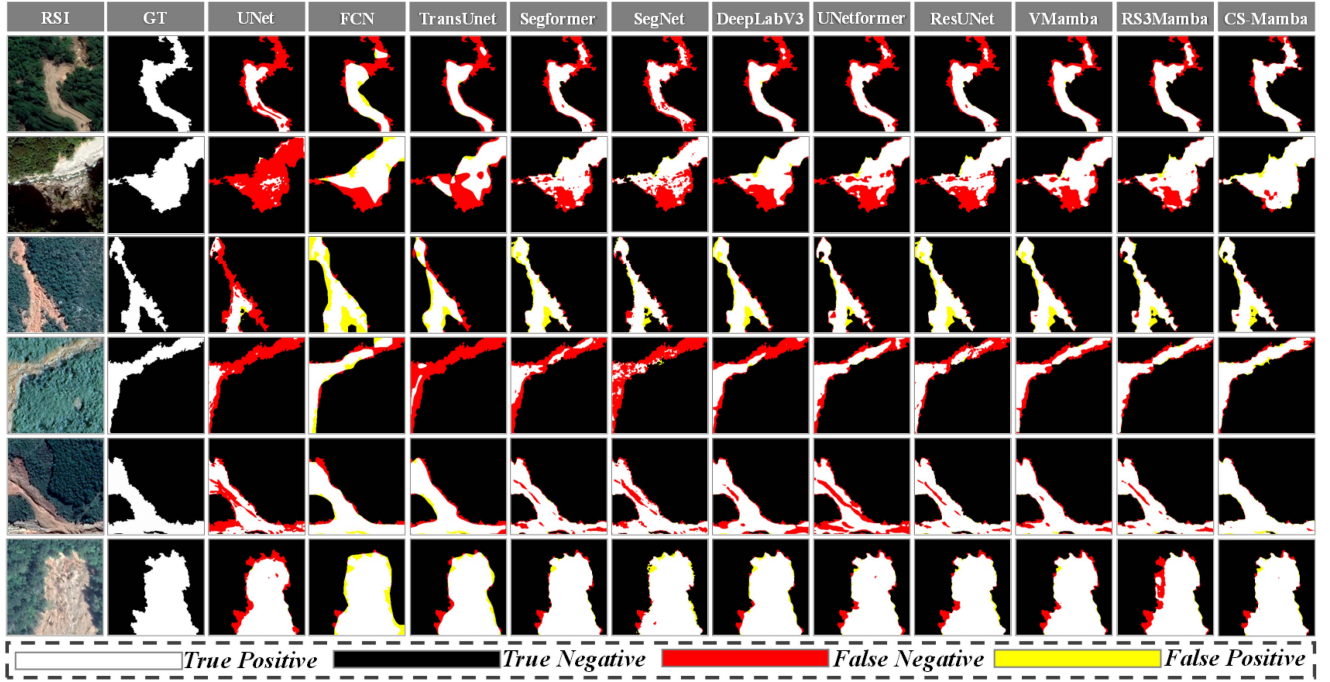


Fig. 9. Visual comparison of landslide detection results produced by different methods on GVLM dataset.

detection results reveal that, although models such as Segformer and RS3Mamba achieve relatively good performance in major landslide regions, they often suffer from large missed detection areas (red) or false detection areas (yellow) near landslide edges and fault zones. CS-Mamba produces more complete landslide contours, with significantly reduced false and missed detection areas, and boundaries that closely match the labels. This advantage can be attributed to the model's improved strategy for integrating global and local information, enabling precise detection and strong suppression of nonlandslide objects in complex backgrounds, thus enhancing spatial consistency and reliability.

On the Bijie dataset, CS-Mamba also maintains leading detection performance, with an mIoU of 86.18% and an F1 score of 92.15%. As shown in Fig. 10, the results highlight its superior capability in handling fragmented landslides, small targets, and regions with boundary adhesion. Compared with architectures such as UNetformer and Vmamba, the proposed method effectively reduces the yellow false detection areas and red missed detection bands, producing smooth boundaries that closely align with the ground truth contours. In complex terrains and under extreme conditions, CS-Mamba demonstrates noticeably higher detection accuracy for small patches and debris slides, showing strong adaptability to fine-grained features and enhanced spatial representation capability.

Overall, CS-Mamba achieves superior performance across three landslide datasets. It excels at detecting landslide boundaries across varying scales and irregular shapes, significantly reducing missed detections. In challenging environments, the model exhibits strong discriminative ability, effectively suppressing false alarms in scenarios with unclear boundaries or target adhesion, thereby improving the overall detection accuracy.

TABLE III
ABLATION EXPERIMENTS ON GVLM DATASET

	Exp ID	Precision	Recall	OA	MIoU	F1
B: Full VSSM						
C: Cross Embedding Module						
M: Multi-scale Aware VSSM						
S: Semantic Fusion Module						
B	A1	92.44	91.83	97.61	86.10	92.13
B+C	A2	93.12	92.74	98.48	87.43	92.93
B+C+M	A3	93.02	93.89	97.16	88.10	93.45
B+M	A4	92.99	93.21	97.44	87.58	93.10
B+M+S	A5	93.81	94.28	97.88	89.14	94.04
B+C+M+S	A6	95.31	95.05	98.88	90.90	95.18

C. Ablation Experiment

To evaluate the contribution of each component in CS-Mamba, a series of ablation experiments were conducted on the GVLM dataset under the default experimental settings. In these experiments, the main backbone components of CS-Mamba were replaced with VSSM blocks to construct a baseline model referred to as the full VSSM architecture. The CEM, MAVSS, and SFM modules were then introduced sequentially to quantify their individual contributions to the model's performance. Table III gives the detection results for different combinations of these modules.

Starting from the baseline model, we progressively incorporate the CEM, MAVSS, and SFM modules, where the SFM essentially acts as a fusion unit that requires two inputs: the local features from the encoder layers and the global semantics from the MAVSS. The CEM (A2) significantly improves OA (+0.87%) and mIoU (+1.33%), indicating that explicit

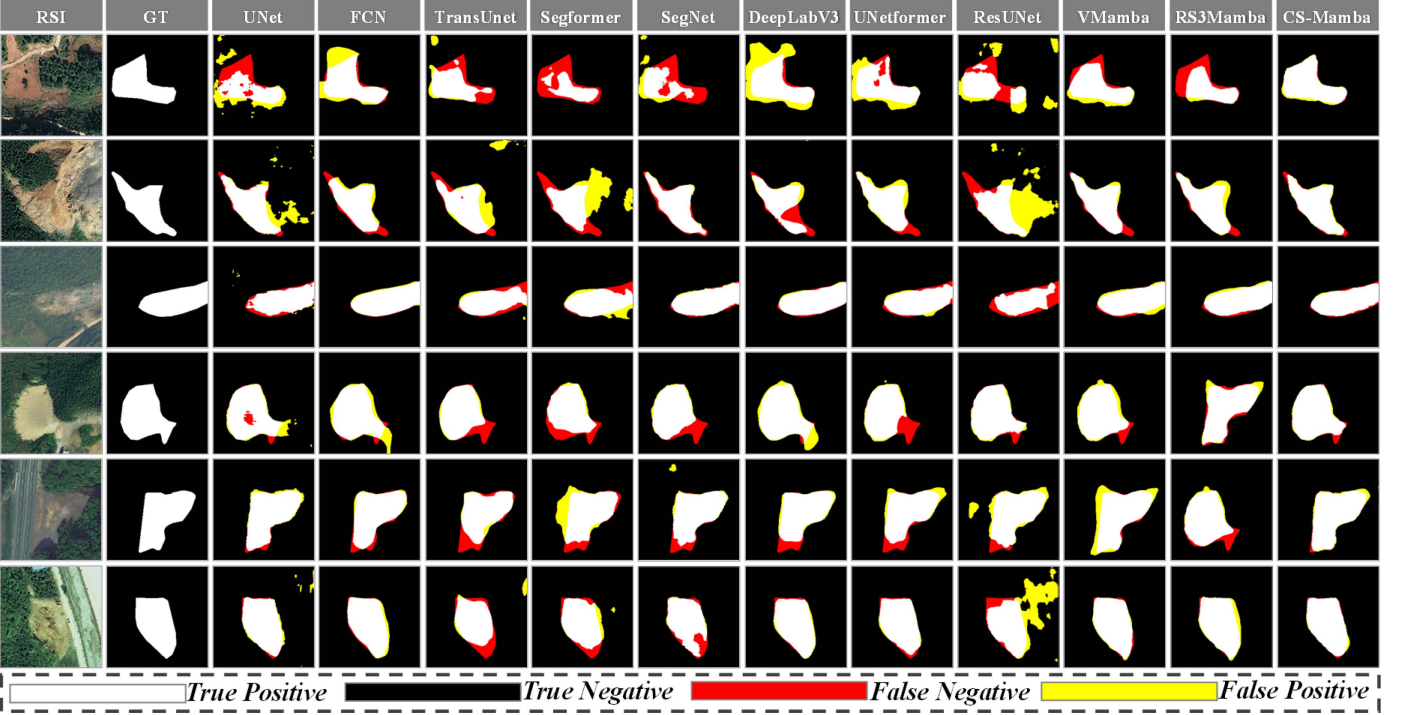


Fig. 10. Visual comparison of landslide detection results produced by different methods on Bijie dataset.

multiscale embedding effectively enhances the model's ability to capture diverse landslide contexts. This improvement suggests that aligning features across different spatial scales helps reduce semantic inconsistency caused by large variations in landslide size.

Further introducing the MAVSS (A3) enhances both recall and mIoU, demonstrating the effectiveness of jointly modeling global contextual dependencies and fine-grained local details. This result confirms that MAVSS plays a key role in alleviating scale-induced feature fragmentation and enhancing semantic coherence. A comparison between A3 and A4 shows that although the standalone use of MAVSS (A4) yields some improvements over the baseline, its combination with the CEM (A3) substantially boosts segmentation performance, reflecting their complementary strengths. Incorporating the SFM (A5) leads to a considerable increase in both F1 score and mIoU, underscoring the importance of adaptive semantic fusion in refining ambiguous regions and improving boundary delineation. Finally, the complete model (A6) achieves the best performance across all metrics, with particularly notable gains in mIoU (+4.80%) and F1 (+3.05%) compared with the baseline, validating the synergistic benefits of integrating all three modules.

To further verify the practical contribution of each module to the model's perceptual capability, we conducted a visualization analysis of the intermediate feature maps for key module combinations (A1, A2, A3, A6). As shown in Fig. 11, the baseline model (A1) fails to fully capture the spatial extent of the landslide regions, with weak and fragmented feature responses. After introducing the CEM (A2), the depiction of boundaries improves, and background noise is effectively suppressed. With the further addition of the MAVSS (A3), the activation responses

over the landslide regions become more coherent, filling the gaps between responses and improving semantic consistency. The complete model (A6) exhibits the clearest and most spatially complete feature responses, with sharp boundaries and a markedly reduced false detection rate, fully demonstrating the complementary role of multiscale context modeling (M) and the SFM (S) in the feature extraction process.

D. Model Complexity Analysis

In addition to demonstrating superior detection performance across multiple landslide datasets, the computational complexity of the model is an important factor that must be considered for practical applications. The number of floating-point operations (FLOPs) and the memory footprint directly influence its potential for deployment in large-scale and real-time remote sensing scenarios. FLOPs measure the computational complexity of the model, whereas the memory footprint evaluates the network's memory requirements. An ideal model should maintain low values in both FLOPs and memory consumption.

As given in Table IV, compared with other Mamba-series models with input size of 512×512 and batch size = 1, the proposed CS-Mamba significantly reduces computational complexity while maintaining reasonable memory usage. The stacking of MAVSS blocks in the bottleneck introduces some additional complexity relative to lightweight models such as UNetformer and Segformer. Nevertheless, it is worth emphasizing that CS-Mamba achieves the highest mIoU of 90.90% while keeping computational costs low, indicating that the proposed architecture attains superior segmentation performance while effectively controlling complexity.

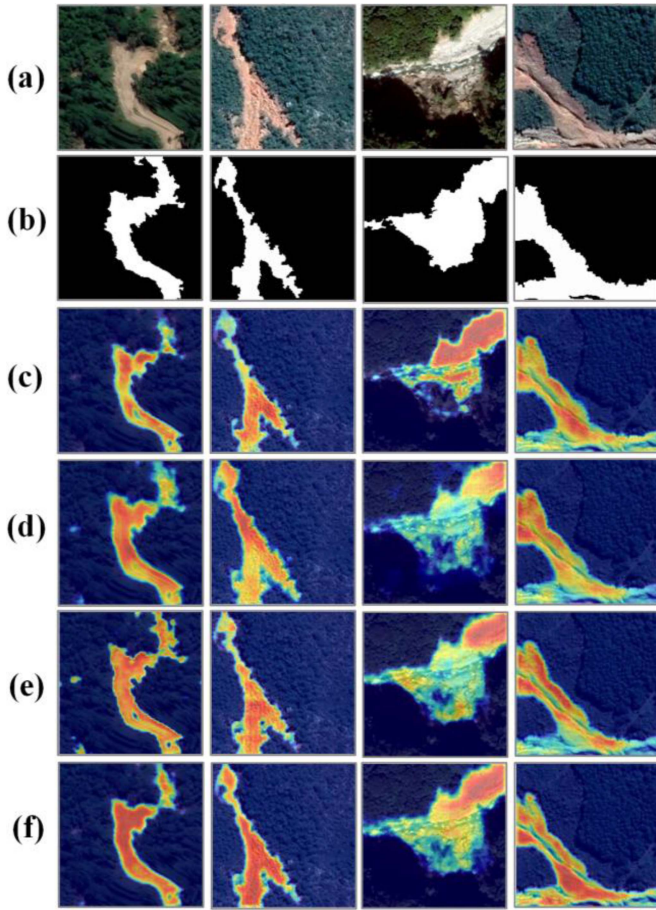


Fig. 11. Feature map visualization results from ablation experiment. (a) Images. (b) GT. (c) Feature map from experiment A1. (d) Feature map from Experiment A2. (e) Feature map from experiment A3. (f) feature map from experiment A6.

TABLE IV
COMPARISON OF MODEL COMPLEXITY OF DIFFERENT METHODS

Model	FLOPs(G)	Memory(MB)	mIoU
FCN	50.29	249.97	89.68
UNet	334.27	465.30	87.49
DeepLabV3+	63.94	229.34	89.07
SegNet	321.35	843.15	89.25
ResUNet	230.39	607.05	88.96
UNetformer	23.48	197.90	90.35
TransUNet	234.02	484.90	87.59
Segformer	26.45	269.52	90.18
VMamba	285.4	296.14	88.52
RS3Mamba	202.81	473.08	88.25
Ours	86.53	441.81	90.90

The bold values indicate the best performance achieved by the model under the corresponding metric.

Furthermore, to verify the scalability and resource efficiency of CS-Mamba under different input sizes, we compares the GFLOPs and memory usage of several architectures: RS3Mamba; VMamba; ResUNet; Segformer; and CS-Mamba when processing images with sizes ranging from 128 to 2048 pixels. As shown in Fig. 12(a), the computational cost of all models increases exponentially with input size. In medium to high resolution settings (512–2048), CS-Mamba achieves a substantial reduction in FLOPs compared with RS3Mamba, VMamba, and ResUNet, reflecting a more efficient structural design. Although its computation is slightly higher than that of Segformer, the overhead is acceptable given the superior semantic modeling capability. Fig. 12(b) presents the memory usage trends, indicating that CS-Mamba consistently maintains memory consumption within a reasonable range for input sizes between 128 and 1024, significantly lower than ResUNet and comparable to RS3Mamba and VMamba. At the maximum input size of 2048×2048 , CS-Mamba's memory usage remains below 4 GB. Although there is an increase at this scale, it is still lower than that of the other four models, confirming the proposed model's strong scalability and adaptability for deployment.

E. Domain Generalization Analysis

To evaluate the generalization capability of CS-Mamba to heterogeneous data from unseen domains, we designed a few-shot adaptation under domain shift experiment, which measures the ability of a model to rapidly recover or improve performance using a small amount of target-domain data. Specifically, representative models that achieved strong performance in earlier experiments were selected, including ResUNet, SegNet, UNetformer, RS3Mamba, and our proposed CS-Mamba. All models were first pre-trained on the LMHLD dataset, then fine-tuned on 1%, 5%, 10%, and 20% of the GVLM dataset, and finally tested in a new region with limited labeled samples.

As shown in Fig. 13, the proposed model consistently ranks among the top across all evaluation metrics in both zero-shot and few-shot settings. Specifically, CS-Mamba achieves the highest mIoU and F1 score without using any target-domain samples, demonstrating strong domain generalization capability. With only 1% of labeled GVLM data, it exhibits a remarkable performance gain, surpassing all competing methods. This trend continues as more data is provided, reflecting the model's superior adaptability and sample efficiency. This strong generalization performance can be attributed to the coordinated design of the proposed modules. The CEM reduces scale-induced semantic discrepancies by aligning features across different spatial resolutions, which is particularly beneficial when transferring between datasets with heterogeneous imaging conditions. The MAVSS block further enhances robustness by jointly modeling global contextual dependencies and local structural information, preventing the model from overfitting to domain-specific textures or appearance patterns. In addition, the SFM adaptively injects global semantic guidance into local feature maps, which helps stabilize boundary predictions and suppress noise in unseen domains. In contrast, models such as SegNet and ResUNet show slower recovery rates and larger performance variations. These

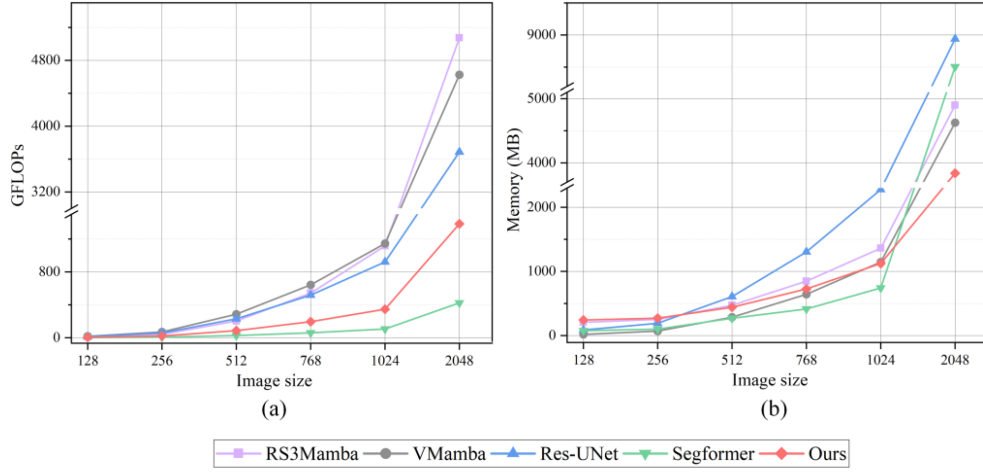


Fig. 12. Computational complexity comparison of different models at different image resolutions. (a) Comparison in terms of GFLOPs for different input sizes. (b) Comparison in terms of memory usage for different input sizes.

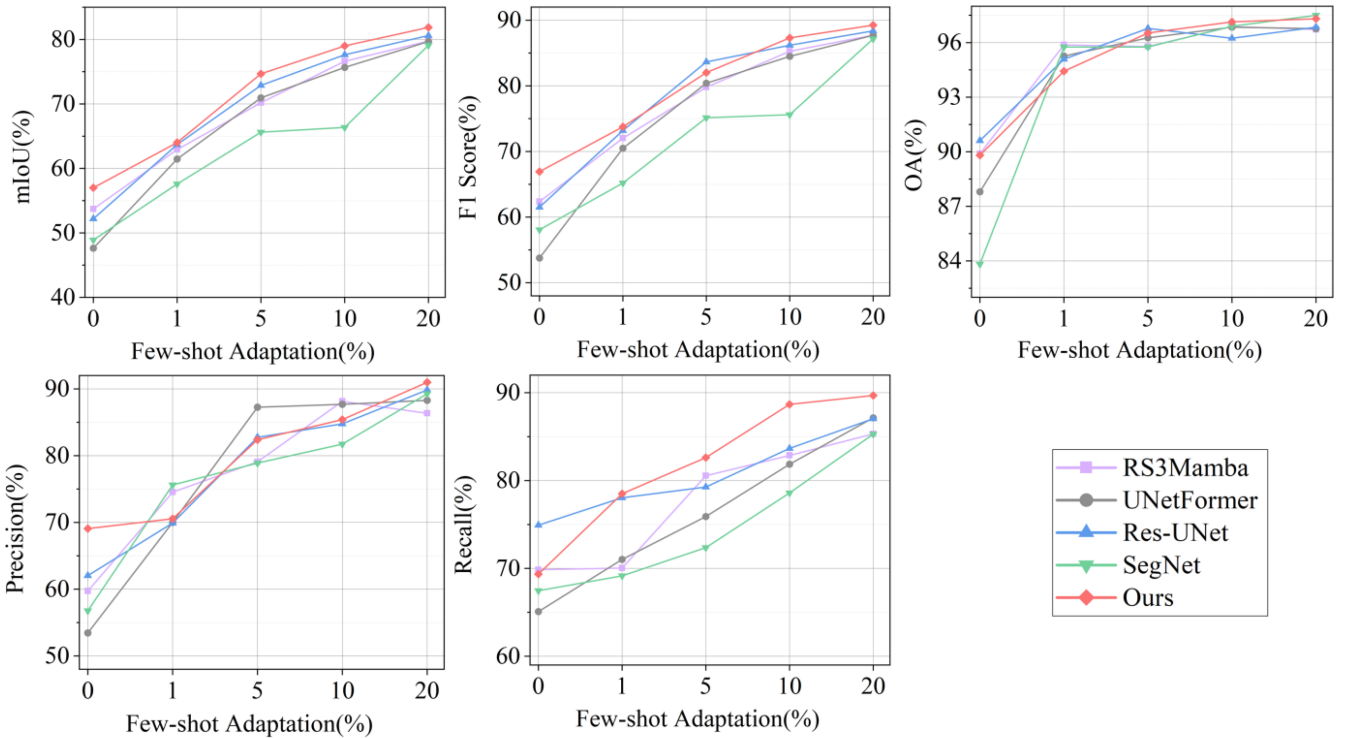


Fig. 13. Comparison of domain generalization and few-shot adaptation performance across different models.

results suggest that the proposed model is well-suited for deployment in scenarios where labeled data are scarce or expensive to obtain, such as landslide monitoring in remote regions.

V. DISCUSSION

A. Advantages and Contributions

The comprehensive experimental results presented in Section IV demonstrate that the proposed CS-Mamba achieves strong performance in high-resolution optical imagery landslide detection. Unlike conventional CNN, Transformer, or Mamba

approaches that typically model features at a fixed scale, CS-Mamba explicitly models multiscale feature consistency and cross-scale semantic alignment, which is critical for accurately delineating complex landslide boundaries. The framework integrates three key modules: the CEM for parallel multiscale convolutional encoding; the MAVSS block for bidirectional interlayer semantic flow; and the SFM for adaptive global-local semantic integration.

Extensive experiments were conducted on three representative landslide datasets—LMHLD, GVLM, and Bijie. CS-Mamba achieved the highest mIoU of 87.11%, 90.90%, and

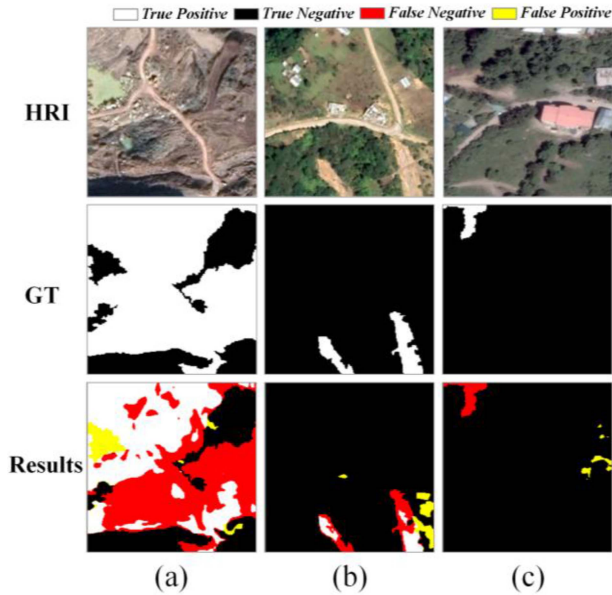


Fig. 14. Examples for challenges in landslide detection from high resolution imagery. (a) Large-scale morphologically irregular landslides with high spectral similarity to bare ground. (b) Small-scale landslides missed during downsampling. (c) Man-made structures misclassified as landslides.

86.18% on the respective datasets, surpassing other state-of-the-art models by margins of 0.45% to 15.87% in mIoU. The method also maintains relatively low computational costs, indicating strong deployment efficiency. Furthermore, CS-Mamba exhibits excellent cross-domain generalization. Under a zero-shot setting without any target-domain data, the model achieves an mIoU of 57% on the GVLM dataset. When only 10% of the target-domain samples are used for fine-tuning, CS-Mamba surpasses all competing models. These results demonstrate strong adaptability in data-scarce scenarios.

B. Limitations and Future Perspectives

Despite the overall strong performance of CS-Mamba, several limitations remain, particularly in challenging scenarios such as small-scale landslides and densely vegetated or geomorphologically complex areas. In Fig. 14(a) and (b), large-scale, morphologically irregular landslides exhibit high spectral similarity to bare ground, leading to extensive omission errors; small-scale landslide targets are prone to being overlooked during downsampling, resulting in missed detections. In Fig. 14(c), man-made structures such as buildings, due to their similar textures, are often misclassified as landslides, producing a substantial number of FPs. We attribute these issues to two main causes:

- 1) Most existing datasets adopt fixed-size image patches as model input. While this facilitates batch processing and computational resource management, it can lead to insufficient contextual information in local regions—particularly when handling large-scale landslides or cross-region geomorphology—causing semantic discontinuities and structural fragmentation. During the limited contextual support provided by fixed-size patches, which weakens the discriminative signals of tiny landslides, especially

when they are partially occluded or have low contrast to the background.

- 2) The CS-Mamba relies on high-resolution optical appearance cues, which become unreliable in densely vegetated regions or geomorphologically complex environments. Vegetation canopy and shadows can partially obscure landslide bodies, and mixed pixels may blur the transition between landslide and background, leading to boundary ambiguity and increased omission/commission errors. Under such conditions, even with global-local modeling, the network may fail to recover clear boundary evidence from RGB imagery alone, which limits performance in heavily covered scenes.

In future work, we will focus on addressing these two aspects. Specifically, we plan to explore boundary and detail aware training objectives and dynamic receptive field mechanisms (or high-resolution auxiliary branches) to preserve fine-grained cues that are easily lost during downsampling. Furthermore, incorporating complementary modalities (e.g., SAR, DEM, LiDAR) and vegetation or land-cover priors to provide structure and terrain aware cues that are less sensitive to optical occlusion and investigate hard negative mining and domain-invariant representation learning to reduce FPs in appearance-similar backgrounds.

VI. CONCLUSION

This article proposes a novel semantic segmentation framework, CS-Mamba which specifically designed for landslide detection in high-resolution remote sensing imagery. The primary contribution lies in its architecture, which explicitly models multiscale feature consistency and cross-scale semantic alignment—two critical aspects often overlooked by existing approaches. By introducing the CEM early in the encoding stage, CS-Mamba effectively enhances the model's capability to capture both fine-grained details and broad spatial context, thereby improving detection performance across landslides of varying scales. The MAVSS block incorporates bidirectional information flow between feature hierarchies and integrates global and local features through a gated fusion mechanism, boosting boundary delineation precision and semantic coherence. The SFM further bridges the gap between global context and local details by dynamically adjusting features across layers, leading to more accurate segmentation of blurred or fragmented landslide regions. Extensive experiments conducted on three diverse datasets—LMHLD, GVLM, and Bijie, demonstrate that CS-Mamba consistently outperforms state-of-the-art methods in both mIoU and F1-score, with particularly notable advantages in small-object segmentation and boundary restoration tasks. In addition, CS-Mamba exhibits strong cross-domain generalization, maintaining high accuracy on unseen data even with minimal fine-tuning. Its relatively low FLOPs and memory consumption further validate its practicality for real-world deployment.

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